JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON SUNDAY 05th APRIL 2026) **TIME : 9:00 AM TO 12:00 NOON**

CHEMISTRY — TEST PAPER WITH SOLUTION

SECTION-A

51. How many grams of residue is obtained by heating 2.76 g of silver carbonate ?

(Given : Molar mass of C, O and Ag are 12, 16 and 108 g mol^{-1} respectively)

(1) 1.08 g (2) 2.16 g

(3) 3.24 g (4) 4.32 g

Sol. $Ag_2CO_3(s) \xrightarrow{\Delta} 2Ag(s) + CO_2(g) + \frac{1}{2}O_2(g)$

2.76g

$\frac{1}{100}$ mol

Moles of Ag(s) = $\frac{2}{100}$ mol

Mass of residue = 2.16 g

52. Arrange the following atomic orbitals of multi electron atoms in order of increasing energy.

(A) n = 3, ℓ = 2, m = +1 (B) n = 4, ℓ = 0, m = 0

(C) n = 6, ℓ = 1, m = 0 (D) n = 5, ℓ = 1, m = +1

(E) n = 2, ℓ = 1, m = +1

Choose the correct answer from the options given below :

(1) C < D < B < A < E (2) B < A < E < C < D

(3) E < C < D < B < A (4) E < B < A < D < C

Sol. (A) n = 3, l = 2, m = +1 ; (n + l) = 5

(B) n = 4, l = 0, m = 0 ; (n + l) = 4

(C) n = 6, l = 1, m = 0 ; (n + l) = 7

(D) n = 5, l = 1, m = +1 ; (n + l) = 6

(E) n = 2, l = 1, m = +1 ; (n + l) = 3

More the value of (n + l), more will be the energy of that particular orbital.

53. Identify the **correct** statements from the following :

A. Heisenberg uncertainty principle is applicable to electron.

B. The size of $2p_x$ orbital is less than the size of $3p_x$ orbital.

C. The energy of 2s orbital of H atom is equal to the energy of 2s orbital of Li.

D. The electronic configuration of Cr is [Ar] $3d^5 4s^1$

Choose the correct answer from the options given below :

(1) A, B and C only (2) A, B and D only

(3) B, C, and D only (4) A, C and D only

Sol. (A) Heisenberg uncertainty principle is applicable to all the moving microscopic particles.

(B) $2p_x$ has less value of n thus has lesser size than $3p_x$.

(C) $E_{2s}(H) \neq E_{2s}(Li)$

(D) Cr :- [Ar] $3d^5$ $4s^1$

Correct statement are A, B and D.

54. What is the mole fraction of water in 10% by weight (w/w) of aqueous urea solution ?

[Given : Molar mass of H, O, C and N are 1, 16, 12 and 14 g mol^{-1} respectively]

(1) 0.825 (2) 0.032

(3) 0.867 (4) 0.967

Sol. Mass ratio = mol ratio × GMM ratio

$\frac{10}{90} = \frac{n}{N} \times \frac{60}{18}$

$\frac{n}{N} = \frac{1}{30}$

$x_{H_2O} = \frac{W}{n+N} = \frac{30}{31} = 0.967$

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55. M_3A_2 is a sparingly soluble salt of molar mass y g mol^{-1} and solubility x g L^{-1}. The ratio of the molar concentration of the anion (A^{3-}) to the solubility product of the salt is

(1) $\frac{1}{54}\cdot\frac{y^4}{x^4}$ (2) $\frac{y^5}{108x^4}$

(3) $108\cdot\frac{x^5}{y^5}$ (4) $\frac{1}{108}\frac{y^4}{x^4}$

Sol.

$$M_3A_2(s) \rightleftharpoons 3M^{+2}(aq.) + 2A^{-3}(aq.)$$

Molarity	$\frac{x}{y}$	–	–
–	3 x/y	2 x/y	

$K_{sp} = [M^{+2}]^3 [A^{-3}]^2$

$K_{sp} = [3x/y]^3 [2x/y]^2$

$$\text{Ratio of } \frac{[A^{-3}]}{K_{sp}} = \frac{2x/y}{\left[\frac{3x}{y}\right]^3\left[\frac{2x}{y}\right]^2}$$

$$= \frac{1}{54}\frac{y^4}{x^4}$$

56. Arrange the following resultant mixtures in increasing order of their pH values

A. 10 mL 0.2 M $Ca(OH)_2$ + 25 mL 0.1 M HCl

B. 10 mL 0.01 M H_2SO_4 + 10 mL 0.01 M $Ca(OH)_2$

C. 10 mL 0.1 M H_2SO_4 + 10 mL 0.1 M KOH

Choose the correct answer from the options given below :

(1) B < C < A (2) C < A < B

(3) C < B < A (4) A < C < B

Sol. (A) $H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O(\ell)$

t = 0 2.5 mmol 4 mmol

t = ∞ 0 1.5 mmol

an final solution, $[OH^-] = \frac{1.5 \text{ mmol}}{35 \text{ ml}} = \frac{1.5}{35}$

$$pOH = -\log\frac{1.5}{35}$$

$$pH = 14 + \log\left(\frac{1.5}{35}\right)$$

(B) $H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O(\ell)$

t = 0 0.2 mmol 0.2 mmol

final pH = 7

(C) $H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O(\ell)$

t = 0 2 mmol 1 mmol

t = ∞ 1 mmol 0

Final pH < 7

So, order of pH values :

(C) < (B) < (A)

57. First order gas phase reaction

$A \rightarrow B + C$

p_i = initial pressure of gas A, p_t = total pressure of the reaction mixture at time t

Expression of rate constant (k) is

(1) $\frac{1}{t}\ln\frac{p_i}{2p_i - p_t}$ (2) $\frac{1}{t}\ln\frac{2p_i}{p_i - p_t}$

(3) $\frac{1}{t}\ln\frac{p_i}{3p_i - 2p_t}$ (4) $\frac{1}{t}\ln\frac{3p_i}{4p_i - p_t}$

Sol.

	A(g)	→ B(g)	+ C(g)
t = 0	P_i	–	–
t = t	$P_i - x$	x	x

As given $(P_i - x) + x + x = P_t$

$x = (P_t - P_i)$

$$K = \frac{1}{t}\ell n\frac{P_i}{P_i - x} = \frac{1}{t}\ell n\frac{P_i}{P_i - (P_t - P_i)}$$

$$K = \frac{1}{t}\ell n\frac{P_i}{2P_i - P_t}$$

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58. **Statement I :** The correct order of electronegativity of fluorine, oxygen nitrogen is F > O > N.

Statement II : The oxidation state of oxygen in OF_2 is +2 and in Na_2O is –2.

In the light of the above statements, choose the **correct** answer from the options given below

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

Sol. Statement-I

$\left.\begin{array}{l}(EN)_F = 4.0\\(EN)_O = 3.5\\(EN)_N = 3.0\end{array}\right\}$ on pauling scale

$(EN)_F > (EN)_O > (EN)_N$

Statement-II

Oxidation state of oxygen in OF_2

$OF_2 \Rightarrow x - 1 \times 2 = 0$

$x = +2$

Oxidation state of oxygen in Na_2O

$Na_2O \Rightarrow +1 \times 2 + x = 0$

$x = -2$

Both the statement I and statement II are correct.

59. Correct statements from the following are :

A. Nitrogen in oxidation states from +1 to + 4 disproportionates in acid medium.

B. Nitrogen has the ability of form dπ-pπ multiple bonds with itself and other elements with small size and high electronegativity.

C. N-N single bond is stronger than P-P single bond.

D. Nitrogen has highest density in its group due to small size.

E. The maximum covalency of nitrogen is four since it has only four valence orbitals for bonding.

Choose the correct answer from the options given below :

(1) B, C, and D only (2) C, D, and E only

(3) A, C and E only (4) A and E only

Sol. (A) In the case of nitrogen, all oxidation states from + 1 to +4 tend to disproportionate in acid solution. For example.

$3HNO_2 \rightarrow HNO_3 + H_2O + 2NO$

(B) Nitrogen form $p_\pi - p_\pi$ bonds with itself and other elements with small size and high electronegativity, $d_\pi - p_\pi$ is not possible with itself as it has no d-orbital.

(C) The single N-N bond is weaker than the single P–P bond because of high intereelecronic repulsion of the non-bonding electrons (lone pair), owing to the small bond length.

(D) Nitrogen has lowest density in its group.

(E) The maximum covalency of nitrogen is four.

Correct answer are (A) and (E)

60. Which of the following is NOT a physical or chemical characteristics of interstitial compounds ?

(1) They have high melting points, higher than those of pure metals

(2) They are very soft ionic in nature

(3) They retain metallic conductivity

(4) They are chemically inert and usually non-stoichiometric

Sol. The physical and chemical characteristies of interstitial compounds are :

(1) They have high melting points, higher than those of pure metals.

(2) They are very hard, some borides approach diamond in hardness.

(3) They retain metallic conductivity

(4) They are chemically inert and usually non-stoichiometric.

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61. The correct statements about metal carbonyls are

A. The metal-carbon bonds in metals carbonyls possess both σ and π-character.

B. Due to synergic bonding interactions between metals and CO ligand the metal-carbon bond becomes weak.

C. The metal-carbon σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of metal.

D. The metal-carbon π bond is formed by the donation of electrons from filled d-orbital of metal into vacant π* orbital of CO.

Choose the correct answer from the options given below :

(1) A and B only (2) A, C and D only

(3) B and C only (4) A and D only

Sol. (A) The metal-carbon bonds in metal carbonyls possess both σ and π-character.

(B) The metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal i.e. metal carbon bond becomes strong.

(C) The metal-carbon σ is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of metal.

(D) The metal-carbon π bond is formed by the donation of electrons from filled d-orbial of metal into vacant π* orbital of CO.

Correct answer are A, C, D

62. Given below are two statements :

Statement I : Each electron in e_g orbitals destabilizes the orbitals by $+0.6\ \Delta_0$ and each electron in the t_{2g} orbitals stabilizes the orbitals by $-0.4\ \Delta_0$ in an octahedral field on the basis of crystal field theory.

Statement II : All the d-orbitals of the transition metals have the same energy in their free atomic state but when a complex is formed the ligands destroy the degeneracy of these orbitals on the basis of crystal field theory.

In the light of the above statements, choose the correct answer from the options given below.

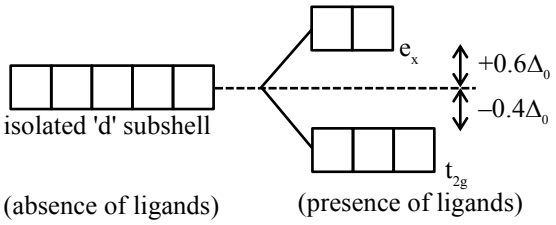
(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

Sol. **Statement-I**



Statement-II

The degeneracy of the d-orbitals is removed due to ligand electron-metal electron repulsions according to crystal field theory.

Both the statement I and II are correct.

63. Given below are two statements :

Statement I : On the basis of inductive effect, the order of stability of alkyl carbanions is $CH_3^- > CH_3 - CH_2^- > (CH_3)_2CH^- > (CH_3)_3C^-$.

Statement II : Allyl and benzyl carbanions are more stabilised by inductive effect and not by resonance effect.

In the light of the above statements choose the correct answer from the options given below

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct and Statement II is incorrect

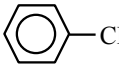
(4) Statement I is incorrect and Statement II is correct

Sol. **Statement-I**

$$\overset{\ominus}{C}H_3 > CH_3 - \overset{\ominus}{C}H_2 > (CH_3)_2\overset{\ominus}{C}H > (CH_3)_3\overset{\ominus}{C}$$

It is correct order of anion stability and it is based on inductive effect.

Statement-II

$CH_2 = CH - \overset{\ominus}{C}H_2$ & 

Allyl anion

Benzyl carbanion

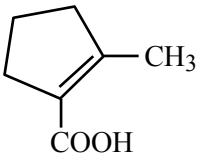
Both are stabilised by resonance

Statement-I : Correct

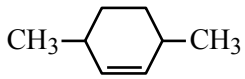
Statement-II : Incorrect

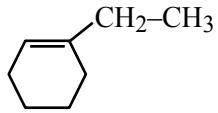
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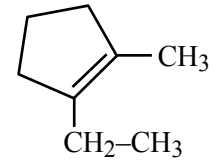
64. "P" is hydrocarbon of molecular formula : C_8H_{14}. On ozonolysis, "P" forms "Q". "Q" on treatment with alkali under reflux condition produces. "R", which on treatment with I_2/NaOH gives a yellow precipitate. Acidification of the solution gives "S". The structure of "S" given below :

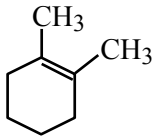


The correct structure of "P" is

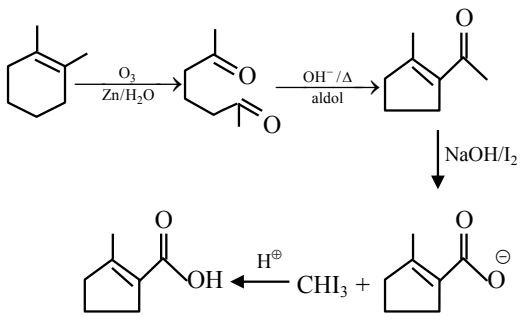
(1) 

(2) 

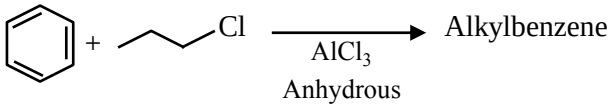
(3) 

(4) 

Sol.



65. For the following Friedel Craft's alkylation reaction, which of the statements are **correct** ?



A. Major product is n-propyl benzene.

B. iso-propyl carbocation intermediate is also generated

C. Multiple substitution is inevitable.

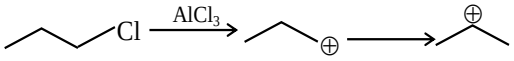
D. Introducing electron-donating substituent on benzene will not produce any alkylbenzene.

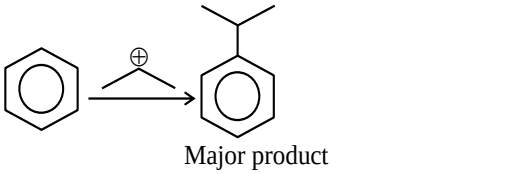
Choose the correct answer from the options given below :

(1) A and D only (2) B and C only

(3) A and C only (4) B and D only

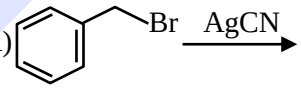
Sol.

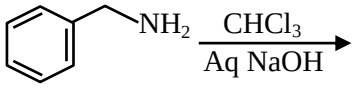


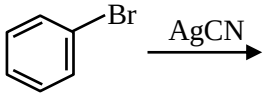


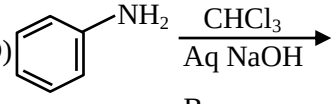
Major product

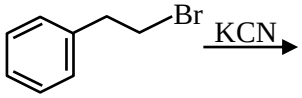
66. Benzyl isocyanide can be obtained from

(A) 

(B) 

(C) 

(D) 

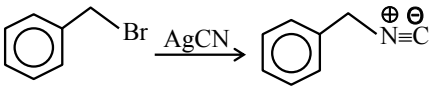
(E) 

Choose the correct answer from the options given below :

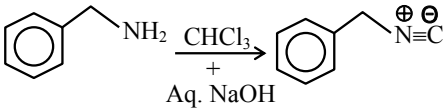
(1) A and B only (2) A and C only

(3) B and D only (4) D and E only

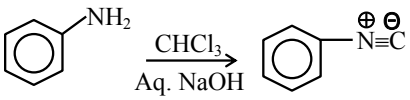
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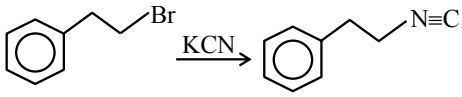
Sol. (A) 

Benzyl Isocyanide

(B) 

(C) 

(D) 

(E) 

(A) & (B) gives correct product.

67. Consider compounds A, B and C with following structural formulae

A = $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - OH$

B = $CH_2 = CH - CH_2 - CH_2 - CH_3$

C = $HO - CH_2 - CH_2 - CH(OH) - CH_3$

For the conversion of B from A, reagent (D) required is _____ and structural formula of product (E) obtained when C undergoes same reaction using excess reagent (D) is _____

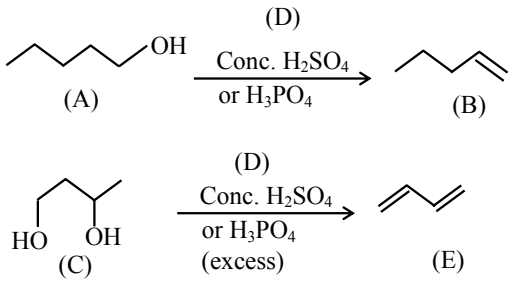
(1)	D	E
	Conc. H_2SO_4	$CH_2 = CH - CH(OH)/CH_3$

(2)	D	E
	PCC	$HO - CH_2 - CH_2 - CH = CH_2$

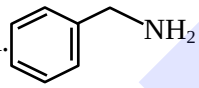
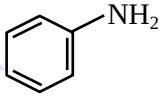
(3)	D	E
	PCC	$CH_2 = CH - CH = CH_2$

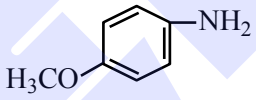
(4)	D	E
	Conc. H_2SO_4 or H_3PO_4	$CH_2 = CH - CH = CH_2$

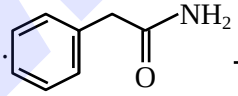
Sol.

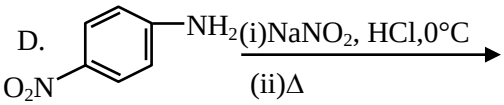


68. Identify the incorrect statements :

A.  is a stronger base than 

B.  can be synthesized by Gabriel phthalimide synthesis.

C.  Primary aromatic amine

D.  product will dissolve in NaOH

Choose the correct answer from the options given below :

(1) A and D only

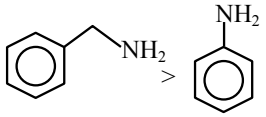
(2) A and C only

(3) B and C only

(4) A and B only

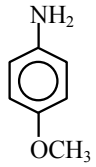
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Sol. (A)



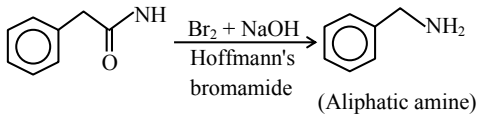
(Basic strength order)

(B)



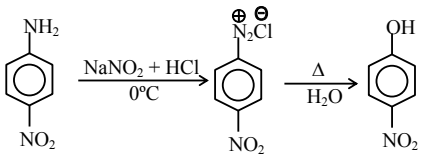
Cannot be synthesized by Gabriel pathalamide synthesis

(C)



(Aliphatic amine)

(D)



Soluble in NaOH

Only (B) and (C) are incorrect hence

69. Identify the correct statements

A. Glucose exist in two anomeric forms.

B. Anomers of glucose differ in configuration at C-1 in cyclic hemiacetal structure.

C. Melting point of α-anomer of glucose is greater than β-anomer.

D. Specific rotation of α-anomer is +19° while for β-anomer is +112°

E. α and β-anomers of glucose are prepared by crystallization of saturated glucose solution at 303 K and 371 K respectively.

Choose the correct answer from the options given below :

(1) A and B only

(2) B and C only

(3) A, B and D only

(4) A, B and E only

Sol. It is based on theory

70. **Statement I :** Sodium dichromate and potassium dichromate are classified as primary standard in titrimetric analysis .

Statement II : Phenolphthalein is a weak base, therefore it dissociates in acidic medium.

In the light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

Sol. $Na_2Cr_2O_7$ is hygroscopic in nature and is not used as primary standard.

Phenolphthalein is weak acid.

SECTION-B

71. Consider the following species

$BrF_5, XeF_5^-, BF_4^-, ICl_4^-, XeF_4, SF_4, NH_4^+, ClF_3, XeF_2, ICl_2^-$

species having sp^3d hybridized central atom is _____

Sol.

Species	Hybridisation
(1) BrF_5	sp^3d^2
(2) $XeF_5^{\ominus}$	sp^3d^3
(3) $BF_4^{\ominus}$	sp^3
(4) $ICl_4^{\ominus}$	sp^3d^2
(5) XeF_4	sp^3d^2
(6) SF_4	sp^3d
(7) NH_4^+	sp^3
(8) ClF_3	sp^3d
(9) XeF_2	sp^3d
(10) $ICl_2^{\ominus}$	sp^3d

Number of species having sp^3d hybridisation are 4.

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72. In an estimation of sulphur by Carius method 0.2g of the substance gave 0.6 g of $BaSO_4$. The percentage of sulphur in the substance is _____%.
(Given molar mass in g mol^{-1} S : 32, $BaSO_4$: 231)

Sol. Moles of $BaSO_4$ formed $\xrightarrow[231gm]{0.6gm}$ = 0.0026

Moles of $BaSO_4$ = Moles of S

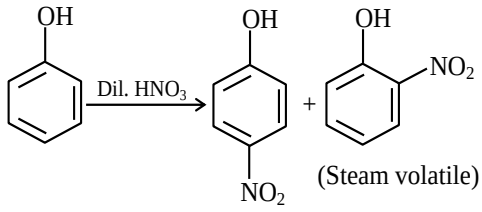
Mass of sulphur (W_S) = $\xrightarrow{\frac{0.6}{231}} \times 32$

% of sulphur (S) = $\frac{0.6\times32}{231\times0.2}\times100$

= $\frac{1920}{46.2} = 41.558 \approx 42$

73. One mole of phenol is treated with dilute HNO_3 at 298K to give a mixture of products. The mixture is separated by steam distillation. The seam volatile compound (X) is separated. The increase in percentage of oxygen in (X) with respect to phenol is _____10^{-1}%
(Given molar mass in g mol^{-1} H:1, C:12, N:14, O:16)

Sol.



(Steam volatile)

% oxygen in phenol = $\frac{16}{94}\times100$ = 17.02%

% oxygen in o-Nitrophenol ($C_6H_5NO_3$)

Molecular mass = $C_6H_5NO_3$ = 139 g/m

= $\frac{48}{139} \times 100 = 34.53\%$

% increase = (34.53 – 17.02) = 17.5

17.5 × 10^{-1} = 175 Answer

74. The values of pressure equilibrium constant recorded at different temperatures for the following equilibrium reaction have been given below:

$A(g) \rightleftharpoons B(g) + C(g)$

$\frac{1}{T}(K^{-1})$	$Log_{10}K_p$
0.05	3.5
0.06	2.5
0.07	1.5

The magnitude of $\frac{\Delta H^\circ}{R}$ calculated from the above data is _____.(Nearest integer)

Sol. Using formula

$\log K_{P_2} - \log K_{P_1} = \frac{\Delta H}{2.303\,R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$3.5 - 2.5 = \frac{\Delta H}{2.303\,R}(0.06 - 0.05)$

$\frac{\Delta H}{R} = \frac{2.303}{0.01}$

$\frac{\Delta H}{R} = 230.3 \approx 230$

75. If the half life of a first order reaction is 6.93 minutes then the time required for completion of 99% of the reaction will be_____minutes.
(Given : log 2 = 0.3010)

Sol. $t_{1/2} = \frac{\ell n2}{K} \Rightarrow K_1 = \frac{0.693}{6.93} \Rightarrow 0.1\ min^{-1}$

$t_{99\%} = \frac{1}{K}\ell n\left[\frac{A_0}{A_t}\right]$

$= \frac{1}{0.1}\ell n\left[\frac{100}{1}\right] \Rightarrow \frac{2\ell n10}{0.1}$

$= \frac{2\times2.303}{0.1}$

$\Rightarrow 46.06\ min \approx 46$

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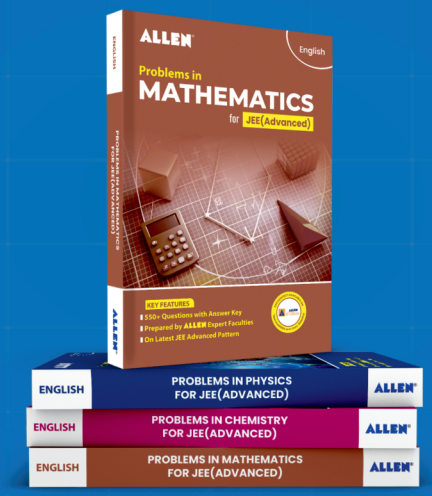


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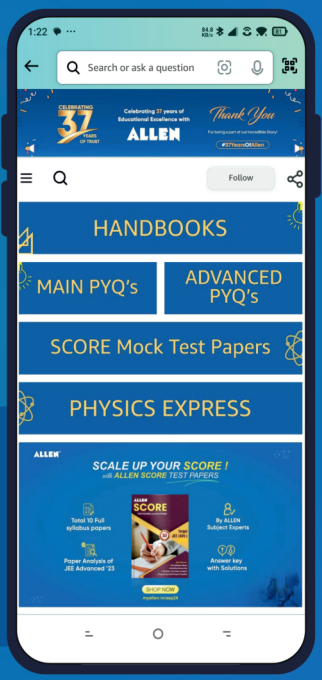
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